Chapter 8

Optimism and Misrepresentation in Early Project Development

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Pervasive misinformation about the costs, benefits and risks involved is a big problem in major project development. Consequences of misinformation are cost overruns, benefit shortfalls and waste. This chapter identifies optimism bias and strategic misrepresentation as main causes of misinformation. Bias and misrepresentation are problems throughout the project cycle, but the problems are greatest during early project development, because here measures to curb bias and misrepresentation are weakest. Measures for improving early project development are presented, emphasizing accountability and better methods.

Misinformation about Costs, Benefits and Risks

Major projects generally have the following characteristics:

- Such projects are inherently risky, due to long planning horizons and complex interfaces.
- Technology and design are often not standard.
- Decision-making and planning are typically multi-player processes with conflicting interests.
- Often there is "lock in" with a certain project concept at an early stage, leaving alternatives analysis weak or absent.
- The project scope or ambition level will typically change significantly over time.
- Statistical evidence shows that such unplanned events are often unaccounted for, leaving budget and time contingencies sorely inadequate.
- As a consequence, misinformation about costs, benefits and risks is the norm.
- The result is cost overruns and/or benefit shortfalls with a majority of projects.

Within this chapter, data are presented on cost overruns and benefit shortfalls for large transportation infrastructure projects. However, comparative studies indicate that the problems identified apply across a wide range of project types, including transportation, public buildings, IT, power plants, dams, water projects, oil and gas extraction, aerospace, defence, development of new major products, plants and markets, and even large mergers and acquisitions (Flyvbjerg, Garbuio and Lovallo, forthcoming).

Table 1 shows the inaccuracy of construction cost estimates measured as the size of cost overrun. The cost study covers 258 projects in 20 nations on five continents. All projects for which data was obtainable were included in the study.[i] Nine out of ten projects in the study have cost overruns. For rail, average cost overrun is 44.7% measured in constant prices. For bridges and tunnels, the equivalent figure is 33.8% and for roads 20.4%. (Flyvbjerg et al, 2002). Overrun is calculated from the time of decision to build (typically the stage of the final business case). If overrun was calculated from earlier stages of project development, i.e. from the draft business case, or pre-feasibility study stages, overrun would typically be substantially higher. The large standard deviations shown in Table 1 are as interesting as the large average overruns. The size of the standard deviations demonstrates that uncertainty and risk regarding cost overruns can be very large. Overrun is found across the nations and is constant for the 70-year period covered by the study. Estimates of cost have not improved over time.

Table 1 Inaccuracy of transportation project cost estimates by type of project, in constant prices.

Type of project	No. of cases (N)	Avg. cost overrun %	Standard deviation
Rail	58	44.7	38.4
Bridges and tunnels	33	33.8	62.4
Road	167	20.4	29.9

Table 2 shows the inaccuracy of travel demand forecasts for rail and road projects. The demand study covers 208 projects in 14 nations on five continents. All projects for which data was obtainable were included in the study.[ii] Nine out of ten rail projects have overestimated traffic, whereas 50% of road traffic forecasts are wrong by more than ±20%; 25% of road traffic forecasts are wrong by more than ±40%. For rail, actual passenger traffic is 51.4% lower than estimated traffic on average. This means that less than half of the passengers forecasted actually showed up on the trains, often resulting in fiscal crises. For roads, actual vehicle traffic is on average 9.5% higher than forecasted traffic. Rail passenger forecasts are highly biased, whereas this is less so for road traffic forecasts. Again the standard deviations are large, indicating a high level of risk and uncertainty. Inaccuracy of forecasts is constant for the 30-year period covered by the study (Flyvbjerg et al, 2005; Flyvbjerg, 2005b).

If techniques and skills for arriving at accurate cost and traffic forecasts have improved over time, these improvements have not resulted in an increase in the accuracy of forecasts.

Table 2 Inaccuracy in forecasts of rail passenger and road vehicle traffic.

Type of project	No. of cases (N)	Avg. inaccuracy %	Standard deviation
Rail	25	–51.4	28.1
Road	183	9.5	44.3

Cost overruns and benefit shortfalls of the frequency and size described above are a problem because they (1) lead to a Pareto-inefficient allocation of resources, i.e. waste; (2) result in non-democratic decisions, because for democracy to work, it must be based on information, not misinformation (3) lead to delays that in turn lead to further cost overruns and benefit shortfalls; (4) destabilize policy, planning, implementation and operations of projects. These problems are getting bigger, because projects are getting bigger.

From the data in Tables 1 and 2, it is clear that when cost and demand forecasts are combined, e.g. in the cost-benefit and environmental impact analyses that are typically used to justify large infrastructure investments, the consequence is inaccuracy to the second degree. Benefit-cost ratios are often wrong, not only by a few per cent, but by several factors. In the worst cases, projects that were projected and promoted to benefit the economy turn out to detract from it instead, producing a negative input to GDP. As a case in point, the Channel Tunnel between France and the UK ended up costing approximately twice that forecast, with only half the revenue. In consequence, Anguera (2006) finds that the actual net present value of the Channel Tunnel to the British economy is negative, namely –£10 billion and that “the British Economy would have been better off had the Tunnel never been constructed.” It is a main task of front-end planning of major projects to avoid this type of situation and to identify projects that will contribute positively to the economy, the environment, etc. This can be done only if it is understood what causes underperforming projects to be chosen for implementation again and again and if this understanding is used to stop bad projects and promote good ones.

There are three main types of explanation that claim to account for inaccuracy in forecasts of costs and benefits in major projects: *technical, psychological and political-economic* explanations.

Technical Explanations

Technical explanations account for cost overruns and benefit shortfalls, in terms of imperfect forecasting techniques, inadequate data, honest mistakes, inherent problems in predicting the future, lack of experience on the part of forecasters, etc. Technical error may be reduced or eliminated by developing better forecasting models, better data and more experienced forecasters. This was, until recently the most common type of explanation of inaccuracy in forecasts (Ascher, 1979; Flyvbjerg et al, 2002 & 2005; Morris and Hough, 1987; Vanston and Vanston, 2004; Wachs, 1990).

However, the credence of technical explanations could mainly be upheld, because samples had been too small to allow tests by statistical methods. The data presented above, which come from the first large-sample study in the field, suggests that technical explanations of forecasting inaccuracy should be rejected, as they do not fit the data well.

Firstly, if misleading forecasts were truly caused by technical inadequacies, simple mistakes and inherent problems with predicting the future, there should be a less biased distribution of errors in forecasts around zero. In fact, for four out of five distributions of forecasting errors, the distributions have a mean statistically different from zero. Only the data for inaccuracy in road traffic forecasts have a statistical distribution that seems to fit with explanations in terms of technical forecasting error.

Secondly, if imperfect techniques, inadequate data and lack of experience were main explanations of inaccuracies, an improvement in accuracy over time might be expected, since in a professional setting, errors and their sources would be recognized and addressed through the refinement of data collection, forecasting methods, etc. Substantial amounts have, in fact, been spent over several decades on improving data and methods. Nevertheless, the data here show that this has had no effect on the accuracy of forecasts. Technical factors, therefore, do not appear to explain the data. It is not forecasting “errors” or their causes that need explaining. It is the fact that, in a large majority of cases, costs are underestimated and benefits overestimated. For technical explanations to be valid, they would have to explain why forecasts are so consistent in ignoring cost and benefit risks, from early project development, through approval, to implementation.

Psychological Explanations

Psychological explanations account for cost overruns and benefit shortfalls in terms of what psychologists call “the planning fallacy and optimism bias”. Such explanations have been developed by Kahneman and Tversky (1979a), Kahneman and Lovallo (1993) and Lovallo and Kahneman (2003).

In the grip of the planning fallacy, planners and project promoters make decisions based on delusional optimism, rather than on a rational weighting of gains, losses and probabilities. They overestimate benefits and underestimate costs. They involuntarily spin scenarios of success and overlook the potential for mistakes and miscalculations. As a result, planners and promoters pursue initiatives that are unlikely to come in on budget or on time, or to even deliver the expected returns. Over-optimism can be traced to cognitive biases, i.e. errors in the way the mind processes information. These biases are thought to be ubiquitous, but their effects can be tempered by simple reality checks, thus reducing the odds that people and organizations will rush blindly into unprofitable investments of money and time.

Psychological explanations fit the data presented above. The existence of optimism bias in planners and promoters would result in actual costs being higher and actual benefits being lower, than those forecasted. Consequently, the existence of optimism bias would be able to account, in whole or in part, for the peculiar bias found in the data. Interestingly, however, when forecasters are asked about causes for forecasting inaccuracies in actual forecasts, they do not mention optimism bias as a main cause of inaccuracy (Flyvbjerg et al, 2005). This could, of course, be because optimism bias is subconscious and thus not reflected by forecasters. After all, there is a large body of experimental evidence for the existence of optimism bias and the planning fallacy (Buehler et al, 1994; Buehler et al 1997; Newby-Clark et al 2002). However, the experimental data are mainly from simple, non-professional settings. This is a problem for psychological explanations, because it remains an open question as to whether they are general and extend beyond such simple settings, to include major projects as well.

Optimism bias would be an important and credible explanation of underestimated costs and overestimated benefits in infrastructure forecasting, if estimates were produced by inexperienced forecasters, i.e, persons who were estimating costs and benefits for the first time and who were thus ignorant of the realities of infrastructure building and were not drawing on the knowledge and skills of more experienced colleagues. Such situations may exist and may explain individual cases of inaccuracy. But, given the fact that in modern society it is a defining characteristic of professional expertise that it is constantly tested (through scientific analysis, critical assessment and peer review) in order to root out bias and error, it seems unlikely that a whole profession of forecasting experts would continue to make the same mistakes decade after decade, instead of learning from their actions. Learning would result in the reduction, if not elimination, of optimism bias and the planning fallacy, which should result in estimates becoming more accurate over time.

However, the data clearly shows that this has not happened. The profession of forecasters would indeed have to be an optimistic, non-professional, group to retain their optimism bias throughout the 70-year period covered by this study for costs and the 30-year period covered for patronage and not realise that they were deceiving themselves and others by underestimating costs and overestimating benefits. This would account for the data, but does not seem a credible explanation. Therefore, on the basis of the data, optimism bias should be rejected as the only, or primary, cause of cost underestimation and benefit overestimation in major projects.

Political-Economic Explanations

Political-economic explanations see planners and promoters as deliberately and strategically overestimating benefits and underestimating costs, when forecasting the outcomes of projects. They do this in order to increase the likelihood that it is their project and not the competitor's, that gains approval and funding. Political-economic explanations have been set out by Flyvbjerg et al (2002 & 2005) and Wachs (1989 &1990).

According to such explanations, planners and promoters purposely spin scenarios of success and gloss over the potential for failure. Again, this results in the pursuit of ventures that are unlikely to come in on budget, or on time, or to deliver the promised benefits. Strategic misrepresentation can be traced to political and organizational pressures, being in competition for scarce funds, or jockeying for position and it is rational in this sense. If a lie can be defined as making a statement intended to deceive others (Bok, 1979; Cliffe et al, 2000), it is clear that deliberate misrepresentation of costs and benefits is lying. Furthermore, lying pays off, or at least political and economic agents believe it does. Where there is political pressure, there is misrepresentation and lying, but misrepresentation and lying can be moderated by measures of accountability.

Political-economic explanations and strategic misrepresentation account well for the systematic underestimation of costs and overestimation of benefits found in the data. A strategic estimate of costs would be low, resulting in cost overrun, whereas a strategic estimate of benefits would be high, resulting in benefit shortfalls. A key question for explanations in terms of strategic misrepresentation, is whether estimates of costs and benefits are intentionally biased to serve the interests of promoters in getting projects started. This question raises the difficult issue of lying. Questions about lying are notoriously hard to answer, because a lie is making a statement intended to deceive others and in order to establish whether or not lying has taken place, one must know the intentions of the participants. If promoters and planners have intentionally skewed estimates of costs and benefits to get a project started, they are unlikely to formally tell researchers or others that this is the case, for legal, economic, moral and other reasons. Despite such problems, two studies exist that succeeded in getting forecasters to talk about strategic misrepresentation (Flyvbjerg and Cowi, 2004; Wachs 1990).

Flyvbjerg and Cowi (2004) interviewed public officials, planners and consultants who had been involved in the development of large UK transportation infrastructure projects. The response of a planner with a local transportation authority typifies how respondents explained the basic mechanism of cost underestimation:

> “You will often as a planner know the real costs. You know that the budget is too low but it is difficult to pass such a message to the counsellors [politicians] and the private actors. They know that high costs reduce the chances of national funding.”

Experienced professionals like the interviewee know that out-turn costs will be higher than estimated costs, but because of political pressure to secure funding for projects, they hold back this knowledge, which is seen as detrimental to the objective of obtaining funding. Similarly, an interviewee explained the basic mechanism of benefit overestimation:

> “The system encourages people to focus on the benefits - because until now there has not been much focus on the quality of risk analysis and the robustness [of projects]. It is therefore important for project promoters to demonstrate all the benefits, also because the project promoters know that their project is up against other projects and competing for scarce resources.”

Competition between projects and authorities creates political and organizational pressures, which in turn create an incentive structure that makes it rational for project promoters to emphasize benefits and de-emphasize costs and risks. A project that looks highly beneficial on paper is more likely to get funded than one that does not.

Specialized private consultancy companies are often engaged to help develop project proposals. In general, the interviewees found that consultants showed high professional standard and integrity. But interviewees also found that consultants appeared to focus on justifying projects, rather than critically scrutinizing them. A project manager explained:

> "Most decent consultants will write off obviously bad projects but there is a grey zone and I think many consultants in reality have an incentive to try to prolong the life of projects which means to get them through the business case. It is in line with their need to make a profit."

The consultants interviewed confirmed that appraisals often focused more on benefits than on costs. But they said this was at the request of clients and that, for specific projects discussed, "there was an incredible rush to see projects realized."

One typical interviewee saw project approval as "passing the test" and precisely summed up the rules of the game as:

> "It's all about passing the test [of project approval]. You are in, when you are in. It means that there is so much focus on showing the project at its best at this stage."

Even though this and the previous quotes are about the approval stage in project development, it is not difficult to imagine strategic behaviour in early project development as well. In fact, the UK study indicates that without counter-incentives, the project development process easily degenerates into a justification process, where a project concept chosen at an early stage, i.e. when uncertainty was highest and is given an increasingly positive presentation through pre-feasibility study, draft business case, final business case and approval. The needs which the project is intended to meet and alternative ways of meeting these needs, are ignored or under-analyzed. The initial concept survives intact, often turns out to be the final choice and is implemented. This is a recipe for disaster, but a recipe that is often followed.

The UK study shows that strong interests and incentives exist in project development to present projects as favourably as possible, i.e. with benefits emphasized and costs and risks de-emphasized. Local authorities, developers and land owners, local trade unions, politicians, officials, MPs and consultants all stand to benefit from a project that looks favourable on paper and they have little incentive to actively avoid bias in estimates of benefits, costs and risks. National bodies, e.g. some sections of the Department for Transport and the Ministry of Finance, who fund and oversee projects, may have an interest in more realistic appraisals, but so far they have had little success in achieving such realism. The situation may, however, be changing, with the initiatives to curb bias set out in HM Treasury (2003) and UK Department for Transport (2006).

Wachs (1986 & 1990) found similar results for transit planning in the USA. Taken together, the UK and US studies account well for existing data on cost underestimation and benefit overestimation. Both studies falsify the notion that in situations with high political and organizational pressure, the underestimation of costs and overestimation of benefits is caused by non-intentional technical error, or optimism bias. Both studies support the view that in such situations, promoters and forecasters intentionally use the following formula in order to secure approval and funding for their projects:

Underestimated costs
+ Overestimated benefits
= Project approval

Using this formula and thus "showing the project at its best", as one interviewee said above, results in an inverted Darwinism, i.e. the "survival of the un-fittest." It is not the best projects that get implemented, but the projects that look best on paper. And the projects that look best on paper are the projects with the largest cost underestimates and benefit overestimates, all things being equal. But these are the worst, or "un-fittest," projects, in the sense that they are the very projects that will encounter most problems during construction and operations, in terms of the largest cost overruns, benefit shortfalls and risks of non-viability.

Reference Class Forecasting

When contemplating what planners can do to improve project development, there is a need to distinguish between two different situations: (1) planners and promoters consider it important to obtain accurate forecasts of costs, benefits and risks; and (2) planners and promoters do not consider it important to get forecasts right, because optimistic forecasts are seen as a necessary means of getting projects started. The first situation is the easier one to deal with and here better methodology will go a long way to improve planning and decision-making. The second situation is more difficult. Here changed incentives are essential in order to reward honesty and punish deception: today's incentives often do the exact opposite.

Thus two main measures of improvement are (1) better forecasting methods; and (2) improved incentive structures, the latter being the more important. Better forecasting methods are covered in this section, better incentives in the next.

If planners genuinely consider it important to obtain accurate forecasts, it is recommended they use a new forecasting method, "reference class forecasting", to reduce inaccuracy and bias. This method was originally developed to compensate for the type of cognitive bias in human forecasting that Princeton psychologist Daniel Kahneman found in his Nobel prize-winning work on economic forecasting and decision making (Kahneman, 1994; Kahneman and Tversky, 1979a). Reference class forecasting has proved to be more accurate than conventional forecasting and the method has been endorsed by the American Planning Association (2005):

> "APA [the American Planning Association] encourages planners to use reference class forecasting in addition to traditional methods as a way to improve accuracy. The reference class forecasting method is beneficial for non-routine projects such as stadiums, museums, exhibit centres and other local one-off projects. Planners should never rely solely on civil engineering technology as a way to generate project forecasts."

For reasons of space, only an outline of the method is presented here, based mainly on Lovallo and Kahneman (2003) and Flyvbjerg (2006).

Reference class forecasting consists in taking an "outside view" on the particular project being forecast. The outside view is established on the basis of information from a class of similar projects. The outside view does not try to forecast the specific uncertain events that will affect the particular project, but instead places the project in a statistical distribution of outcomes from this class of reference projects. Reference class forecasting requires the following three steps for the individual project:

(1) Identify a relevant reference class of past projects. The class must be broad enough to be statistically meaningful, but narrow enough to be truly comparable with the specific project.

(2) Establish a probability distribution for the selected reference class.
This requires access to credible, empirical data for a sufficient number of projects within the reference class to make statistically meaningful conclusions.

(3) Compare the specific project with the reference class distribution, in order to establish the most likely outcome for the specific project.

Daniel Kahneman relates the following story about curriculum planning, to illustrate reference class forecasting in practice (Lovallo and Kahneman 2003). Some years ago, Kahneman was involved in a project to develop a curriculum for a new subject area for high schools in Israel. The project was carried out by a team of academics and teachers, who began to discuss how long the project would take to complete. Everyone on the team was asked to write on a slip of paper the number of months needed to finish and report the project. The estimates ranged from 18 to 30 months. One of the team members - a distinguished expert in curriculum development - was then posed a challenge by another team member to recall as many projects as possible that were similar to theirs, and remember these projects as they were at a similar stage to their own project. "How long did it take them from that point to reach completion?" the expert was asked. After a while he answered, with some discomfort, that not all the comparable teams actually completed their task. About 40% of them eventually gave up. Of those remaining, the expert could not think of any that completed their task in less than seven years, or more than ten years. The expert was then asked if he had reason to believe that the present team was more skilled in curriculum development than the earlier ones had been. The expert said no, he did not see any relevant factor that distinguished this team favourably from the teams he had been remembering. His impression was that the present team was slightly below average in terms of resources and potential. The wise decision at this point would probably, according to Kahneman, have been for the team to break up. Instead, the members ignored the pessimistic information and proceeded with the project. They finally completed the project eight years later, and their efforts were largely wasted - the resulting curriculum was rarely used.

In this example, the curriculum expert made two forecasts for the same problem and arrived at very different answers. The first forecast was the inside view; the second was the outside view, or the reference class forecast. The inside view is the one that the expert and other team members adopted. They made forecasts by focusing tightly on the case in hand, considering its objective, the resources they brought to it and the obstacles to its completion. They constructed in their minds scenarios of anticipated progress and extrapolated current trends into the future. The resulting forecasts, even the most conservative, were overly optimistic. The outside view is that provoked by the question put to the curriculum expert. It ignored the details of the project in hand and involved no attempt at forecasting the events that would influence the project's future course. Instead, it examined the experiences of a class of similar projects, laid out a rough distribution of outcomes for this reference class and then positioned the current project in that distribution. The resulting forecast, as it turned out, was much more accurate.

Similarly, to take an example from practical infrastructure planning, city planners preparing to build a new subway would first establish a reference class of comparable projects. This could be the relevant rail projects from the sample used for this article. Through analyses, the planners would establish that the projects included in the reference class were indeed comparable. Secondly, if the planners were concerned with getting construction cost estimates right, they would establish the distribution of outcomes for the reference class regarding the accuracy of construction cost forecasts. Figure 1 shows what this distribution looks like for a reference class relevant to building subways in the UK, developed by Flyvbjerg and Cowi (2004) for the UK Department for Transport. Thirdly, the planners would compare their subway project to the reference class distribution. This would make it clear to the planners that unless they have reason to believe they are substantially better forecasters and planners than their colleagues who did the forecasts and planning for projects in the reference class, they are likely to grossly underestimate construction costs. Finally, planners would then use this knowledge to adjust their forecasts for more realism. Figure 2 shows such adjustments for the UK situation and that, for a forecast of construction costs for a rail project, this forecast would have to be adjusted upwards by 40%, if investors were willing to accept a risk of cost overrun of 50%. If investors were willing to accept a risk of overrun of only 10%, the uplift would have to be 68%. For a rail project initially estimated at, say £4 billion, the uplifts for the 50 and 10% levels of risk of cost overrun would be £1.6 billion and £2.7 billion, respectively.

Figure 1 Inaccuracy of construction cost forecasts for rail projects in reference class. Average cost increase is indicated for non-UK and UK projects separately. Constant prices.

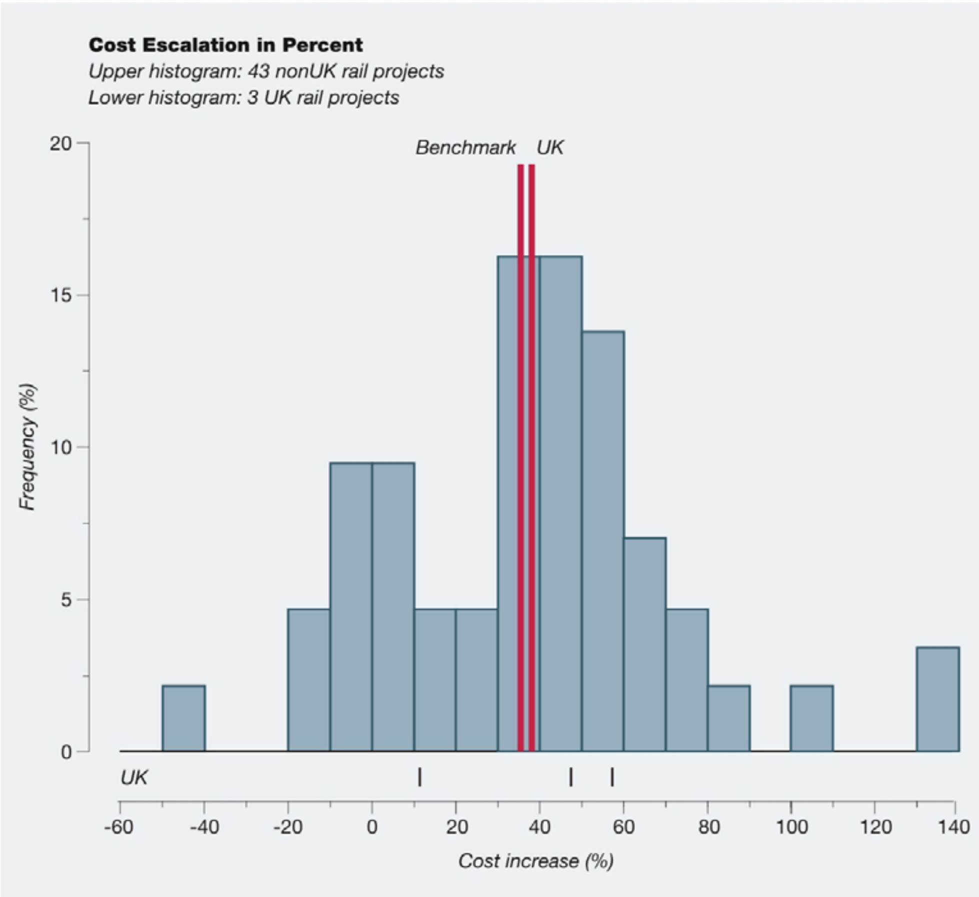
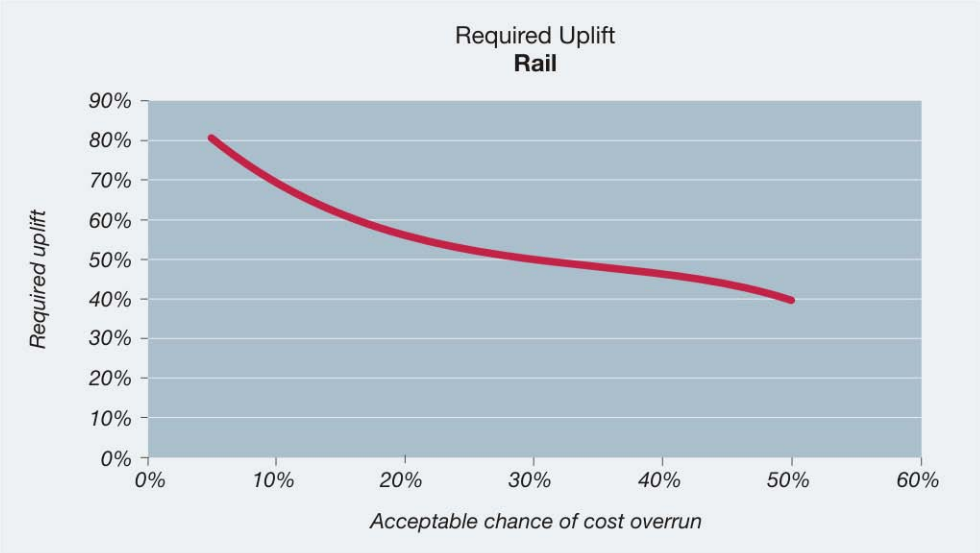


Figure 2 Required adjustments to cost estimates for UK rail projects as function of the maximum acceptable level of risk for cost overrun. Constant prices.



The contrast between inside and outside views has been confirmed by systematic research (Gilovich et al, 2002). The research shows that when people are asked simple questions requiring them to take an outside view, their forecasts become significantly more accurate. However, most individuals and organizations are inclined to adopt the inside view in planning major initiatives. This is the conventional and intuitive approach, for laypeople and experts alike. The traditional way to think about a complex project is to focus on the project itself and its details, to bring to bear what one knows about it, paying special attention to its unique or unusual features and trying to predict the events that will influence its future. The thought of going out and gathering simple statistics about related cases seldom enters a planner's mind. This is usually the case, according to Lovallo and Kahneman (2003). It is certainly the case for cost and benefit forecasting in large infrastructure projects. The first reference class forecast in practical policy and planning was carried out for the Edinburgh Tram in 2004 (Flyvbjerg, 2006).

While understandable, planners' preference for the inside view over the outside view is unfortunate. When both forecasting methods are applied with equal skill, the outside view is much more likely to produce a realistic estimate. This is because it bypasses cognitive and political biases, such as optimism bias and strategic misrepresentation and cuts directly to outcomes. In the outside view, planners and forecasters are not required to make scenarios, imagine events, or gauge their own and others' levels of ability and control, so they cannot get all these things wrong. The outside view, being based on historical precedent, may fail to predict extreme outcomes, i.e. those that lie outside all historical precedents, but, for most projects, it will produce more accurate results. In contrast, a focus on inside details is the road to inaccuracy.

The comparative advantage of the outside view is most pronounced for non-routine projects, that planners and decision-makers in a certain locale have never attempted before, e.g. building an urban rail system in a city for the first time, or a new major bridge, or opera house where none existed before. It is in the planning of such new efforts that the biases toward optimism and strategic misrepresentation are likely to be largest. Choosing the right reference class of comparative past projects may be more difficult when planners are forecasting initiatives for which precedents are not easily found, e.g. the introduction of new and unfamiliar technologies. However, most large infrastructure projects are both non-routine locally and use well-known technologies. Such projects are, therefore, particularly likely to benefit from the outside view and reference class forecasting. The same applies to concert halls, museums, stadiums, exhibition centres and other local one-off projects.

How to Avoid Lock-In and Premature Closure in Early-Stage Development

The typical reference class forecast, as described above, is based upon data on performance for a minimum of 10-15 comparable projects. But for a project in early-stage development, data for this many projects will not often have been collected. Nevertheless, an outside view in early-stage development may help planners both be more realistic from the outset about risks and avoid lock-in. Taking an outside view in early-stage development may be done by carefully selecting a restricted sample of projects and information, which would then constitute a set of "reference points," if not a reference class. The reference points may later form part of a reference class and be used for reference class forecasting,.

Kahneman and Tversky (1979b) argue that the major source of error in planning and forecasting future prospects is the prevalent tendency with planners--be they experts or laypeople--to under weigh or ignore distributional information. Planners "should therefore make every effort to frame the [planning and] forecasting problem so as to facilitate utilizing all the distributional information that is available," say Kahneman and Tversky (1979b: 316). This may be considered the single most important piece of advice regarding how to get better performance in planning and forecasting through improved methods. Incentives should accordingly be established to reward planners who frame planning and forecasting in terms of all available distributional information and to punish planners who ignore such information.

For early-stage planning, this advice, which is firmly grounded in Nobel Prize-winning theories of decision making, is particularly important. This is so because early-stage planning is particularly vulnerable to planners and promoters under weighing and ignoring distributional information. No other stage in the project cycle is more susceptible to premature closure, lock-in, path dependence, anchoring, overconfidence, group think and similar problematic behavior that all result in ignorance of relevant distributional information and thus in inadequate project preparation (Flyvbjerg et al., forthcoming).

Enforcing the outside view is an antidote to such behavior. The outside view may be implemented by reference class forecasting, as we saw above. In early-stage development, where a full reference class forecast is often not possible, a simplified version of the outside view may be implemented by employing maximum variation or extreme case sampling techniques as described by Flyvbjerg (2007). The basic idea is to capture the range of variation that is relevant to the decision at hand and thus to not discard pertinent distributional information.

For instance, an outside view of early-stage development of a given project could be established by collecting data on just two comparable projects, one of which was a significant success and one of which was a significant disaster. This would help early-stage planners get an initial idea of:

(1) the range of risk involved, from success to disaster
(2) the possible causes of success and failure
(3) the potential benefits from considering alternatives.

As a case in point, consider planners and promoters in early-stage deliberation over whether to build a particular example of world-class architecture, e.g. a new opera house, concert hall, museum, or skyscraper designed on the dictum of signature architecture, "build it and they will come." This has recently been the situation for more and more cities, as they jockey for position in an increasingly international and fiercely competitive urban hierarchy. Such planners and promoters could benefit from considering as reference points the extreme cases of the Sydney Opera House and the Guggenheim Museum in Bilbao and from understanding how the first became a complete disaster in terms of delays, cost overruns (a record-setting 1,400 percent) and destroying the career of its architect, Jørn Utzon, while the second, which is no less complex and innovative, was built on time and budget and catapulted its architect, Frank Gehry, into a career of international stardom. (Flyvbjerg 2005a) . Other examples could be used, but the basic idea should be clear: the outside view and a small set of reference points may help establish a much-needed reality check on the project concept in early-stage development, in terms of risk and robustness and thus help prevent lock-in and premature closure.

Public and Private Sector Accountability

This section covers the situation where planners and promoters do not find it important to get forecasts right and therefore do not help to clarify and mitigate risk, but, instead, generate and exacerbate it. Here planners are part of the problem, not the solution. This situation needs some explanation, because it may sound like an unlikely state of affairs. After all, it is generally agreed that planners ought to be interested in being accurate and unbiased in forecasting. The requirement for accuracy is even stated as an explicit requirement in planners' code of ethics, e.g. the code of the American Institute of Certified Planners (AICP), which states that "A planner must strive to provide full, clear and accurate information on planning issues to citizens and governmental decision-makers" (American Planning Association, 1991). The British Royal Town Planning Institute (RTPI, 2001) has laid down similar obligations for its members. The literature is replete with things planners and planning "must" strive to do, but which they do not. Planning must be open and communicative, but is often closed. Planning must be participatory and democratic, but is often an instrument of domination and control. Planning must be about rationality, but is often about power (Flyvbjerg, 1998; Watson, 2003). This is the "dark side" of planning and planners, identified by Flyvbjerg (1996) and Yiftachel (1998) and which is remarkably under-explored by planning researchers and theorists.

Forecasting, too, has its dark side. It is here that "planners lie with numbers," as Wachs (1989) has aptly put it. Planners on the dark side are busy, not with getting forecasts right and following the AICP or RTPI Code of Ethics, but with getting projects funded and built. Accurate forecasts are often perceived to be an ineffective means for achieving this objective. Indeed, accurate forecasts may be seen as counter-productive, whereas false forecasts may be seen as effective in competing for funds and securing the go-ahead for construction. "The most effective planner is sometimes the one who can cloak advocacy in the guise of scientific or technical rationality" (Wachs 1989). Such advocacy would stand in direct opposition to AICP's ruling that "the planner's primary obligation [is] to the public interest" (American Planning Association, 1991).

Nevertheless, seemingly rational forecasts that underestimate costs and overestimate benefits have long been an established formula for project approval and if members of the AICP or RTPI have been excluded for producing inaccurate and biased forecasts, this is seldom broadcast. Enforcement of planners' codes of ethics is lax compared with other professions. If, for instance, medical doctors repeatedly offered diagnoses and prognoses for their patients that were as inaccurate as planners' forecasts of costs and benefits, these doctors would quickly be struck off and out of work. For planners, forecasting is too often not a professional activity, but just another kind of rent-seeking behaviour, resulting in a make-believe world of misrepresentation which makes it extremely difficult to decide which projects deserve undertaking and which do not. The consequence is, as even one of the industry's own bodies, the Oxford-based Major Projects Association, acknowledges, that too many projects go ahead which should not. Conversely, many projects do not proceed that probably should, had they not lost out to projects with "better" misrepresentation (Flyvbjerg et al, 2002).

In this situation, the question is not so much what planners can do, both in early project development and later on, to reduce inaccuracy and risk, but what others can do to impose on planners the checks and balances that might give them the incentive to stop producing biased forecasts and begin to work according to their code of ethics. The challenge is to change the power relations that govern project development and forecasting. Better methods and appeals to ethics are insufficient; institutional change, with a focus on transparency and accountability, is necessary.

Flyvbjerg et al (2003) argue that two basic types of accountability define liberal democracies: (1) public sector accountability through transparency and public control and (2) private sector accountability via competition and market control. Both types of accountability may be effective tools to curb planners' misrepresentation in forecasting and to promote a culture which acknowledges and deals effectively with risk. In order to achieve accountability through *transparency and public control*, the following would be required as practices embedded in the relevant institutions:

- National-level government should not offer discretionary grants to local infrastructure agencies for the sole purpose of building a specific type of infrastructure. Such grants create perverse incentives. Instead, national government should simply offer "infrastructure grants" or "transportation grants" to local governments and let local political officials spend the funds however they choose to, but make sure that every pound they spend on one type of infrastructure reduces their ability to fund another.
- Forecasts should be made subject to independent peer review, e.g. by national or state accounting and auditing offices.
- Forecasts should be benchmarked against comparable forecasts, e.g. reference class forecasting, as described in the previous section.
- Forecasts, peer reviews and benchmarking should be made available to the public as they are produced, including all relevant documentation.
- Public hearings, citizen juries and the like should be organized, to allow stakeholders and civil society to voice criticism and support.
- Scientific and professional conferences should be organized, where planners present and defend their forecasts in the face of colleagues' scrutiny.
- Projects with inflated benefit-cost ratios should be reconsidered and stopped, if re-calculated costs and benefits do not warrant implementation. Projects with realistic estimates of benefits and costs should be rewarded.
- Professional and occasionally even criminal, penalties should be enforced for planners and forecasters who consistently and knowingly produce deceptive forecasts (Garett and Wachs, 1996). Malpractice in planning should be taken as seriously as it is in other professions.

In order to achieve accountability in forecasting via *competition and market control*, the following would be required, again as practices that are both embedded in and enforced by, the relevant institutions:

- The decision to go ahead with a project should, where possible and appropriate, be made contingent on the willingness of private financiers to participate, without a sovereign guarantee for at least one third of the total capital needs. This should be required whether projects are subsidized or not. Private lenders, shareholders and stock market analysts would produce their own forecasts, or would critically monitor existing ones. If they get the forecasts wrong, they and their organizations would be hurt. The result would be more realistic forecasts and reduced risk.

- Forecasters and their organizations must share financial responsibility for covering cost overruns and benefit shortfalls resulting from misrepresentation and bias in forecasting.
- The participation of risk capital should not mean that government gives up or reduces control of the project. On the contrary, it means that government can more effectively play the role it should be playing, namely as the ordinary citizen's guarantor for ensuring concerns about safety, environment, risk and a proper use of public funds.

Whether projects are public, private, or public-private, they should be vested in one and only one, project organization, with a strong governance framework. The project organization may, or may not be a company, may be public or private, or a mixture. What is important is that this organization enforces accountability vis-à-vis contractors, operators, etc. and that, in turn, the directors of the organization are held accountable for any cost overruns, benefits shortfall, faulty designs, unmitigated risks, etc. that may occur during project planning, implementation and operations.

If the institutions with responsibility for developing and building major infrastructure projects were to effectively implement, embed and enforce such measures of accountability, then the misrepresentation in cost, benefit and risk estimates, which is widespread today, could be mitigated. If this is not done, misrepresentation is likely to persist and the allocation of funds for infrastructure to continue to be wasteful and undemocratic.

Clearly, all the measures of accountability mentioned above cannot and will not be in place in early project development. Their implementation is a gradual process that takes place throughout the project cycle. Nevertheless, it is important that measures of accountability are implemented as early as possible and that planners and promoters with early-stage responsibilities know that their actions will eventually be judged by these measures. If accountability is not enforced, the high frequency of project failure documented above is likely to continue.

Signs of Improvement

Fortunately, after decades of widespread mismanagement of major project development, signs of improvement have recently appeared. The conventional consensus, that deception is an acceptable way of getting projects started, is under attack, as will be apparent from the examples below. This is in part because the largest projects are now so big in relation to national economies, that cost overruns, benefit shortfalls and risks from even a single project may destabilize the finances of a whole country or region. This occurred with the 2004 Olympics in Athens and the new international airport in Hong Kong (Flyvbjerg, 2005a). Law-makers and governments begin to see that national fiscal distress is too high a price to pay for the conventional way of developing large projects.

Moreover, with private finance in major projects on the rise, capital funds and banks are increasingly gaining a voice in the project development process. Private investors place their own funds at risk, as opposed to governments, who place the taxpayer's money at risk. Capital funds and banks do not automatically accept at face value the forecasts of project planners and promoters, but typically bring in their own advisers to do independent forecasts and risk assessments - an important step in the right direction.

Finally, democratic governance is generally getting stronger around the world. The Enron scandal and its successors have triggered new legislation and a war on corporate deception that is spilling over into government with the same objective: to curb financial waste and promote good governance. Although progress is slow, good governance is gaining a foothold even in major project development. The main drivers of reform come from outside the agencies and industries conventionally involved in major project development, which increases the likelihood of success.

In 2003 the Treasury of the United Kingdom required, for the first time, that all ministries develop and implement procedures for large public projects, to curb what the Treasury calls "optimism bias." Funding will be unavailable for projects that do not take this bias into account and methods have been developed for how to do this (HM Treasury, 2003; Flyvbjerg and Cowi, 2004; UK Department for Transport, 2006). In the Netherlands, the Parliamentary Committee on Infrastructure Projects conducted extensive public hearings for the first time in 2004. These were to identify measures that would limit the misinformation about large infrastructure projects given to the Parliament, public and media (Tijdelijke Commissie Infrastructuurprojecten, 2004). In Boston, the government has sued to recoup funds from contractor overcharges for the Big Dig related to cost overruns. More countries and cities are likely to follow the lead of the UK, the Netherlands and Boston in coming years; Switzerland and Denmark are already doing so (Swiss Association of Road and Transportation Experts, 2006; Danish Ministry for Transport and Energy, 2006). It is too early to tell whether the measures they implement will ultimately be effective. It seems unlikely, however, that the forces that have triggered the measures will be reversed and it is those forces that reform-minded groups need to support and work with, in order to curb deception and waste. This is the "tension-point" where convention meets reform, power-balances change and new things may happen.

The key weapons in the war on deception and waste are accountability and critical questioning. The professional expertise of planners, engineers, architects, economists and administrators is indispensable in constructing the infrastructures that make society work. However, the claims relating to costs, benefits and risks made by these groups cannot always be trusted and should be carefully examined by independent specialists and organizations. The same applies to claims made by project-promoting politicians and officials. Institutional checks and balances, including financial, professional, or even criminal penalties for consistent and unjustifiable bias in cost estimation, benefits and risks, should be employed. The key principle, often violated today, is that the cost of making a wrong forecast, or a wrong decision, should fall on those making that forecast or decision.

The culture of misinformation in major project development has historical roots which are deeply ingrained in professional and institutional practices. It would be naive to think that this culture is easily toppled. Given the stakes involved - saving taxpayers from billions of dollars of waste, protecting citizens' trust in democracy and the rule of law, avoiding the destruction of spatial and environmental assets - this should not deter us from trying.

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Notes

[i] For details on data and methodology, see Flyvbjerg, Holm and Buhl (2002).
[ii] For details on data and methodology, see Flyvbjerg (2005b).